

18 Negative Binomial Distributions

Example 18.1. Suppose that 86% of Cal Poly students are CA residents. Randomly select Cal Poly students one at a time, independently, until you select 5 CA residents, then stop. Let X be the total number of students selected.

1. Does X have a Binomial distribution? A Geometric distribution? Why or why not?
2. What are the possible values of X ? Is X discrete or continuous?
3. Describe how you could use a tactile simulation to simulate a single value of X .
4. Describe in words how you could use simulation to approximate $P(X = 5)$.
5. Describe in words how you could use simulation to approximate the distribution of X .

6. Compute and interpret $P(X = 5)$.
7. Compute the probability that the first student is not a CA resident but the next 5 are.
8. Why is $P(X = 6)$ different from the probability in the previous part?
9. How many outcomes satisfy the event $\{X = 6\}$? Answer without listing all the possibilities. Hint: careful, it's not $\binom{6}{5}$; why not?
10. Compute and interpret $P(X = 6)$
11. Compute the probability that the first two students are not CA residents but the next 5 are.
12. Why is $P(X = 7)$ different from the probability in the previous part?

13. How many outcomes satisfy the event $\{X = 7\}$? Answer without listing all the possibilities. Hint: careful, it's not $\binom{7}{5}$; why not?
14. Compute and interpret $P(X = 7)$
15. Suggest a formula for the probability mass function of X . Then use the pmf to create a table representing the distribution of X .
16. What seems like a reasonable shortcut formula for $E(X)$?
17. Use the theoretical pmf to compute $E(X)$. Did the shortcut formula work?
18. Interpret $E(X)$ for this example.
19. Compute $\text{Var}(X)$.

20. Would the variance be larger or smaller if we kept selecting until we obtain 10 CA residents instead of 5?

- Perform Bernoulli(p) trials until a fixed number, r , of successes and then stop. Let X count the total number of trials, including the r successes. The distribution of X is defined to be the **Negative Binomial(r, p) distribution**.
- A Negative Binomial distribution is specified by two parameters:
 - r (an integer): the fixed number of successes
 - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
- A discrete random variable X has a **Negative Binomial distribution** with parameters r , a positive integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{x-1}{r-1} p^r (1-p)^{x-r}, \quad x = r, r+1, r+2, \dots$$

- If X has a NegativeBinomial(r, p) distribution

$$\begin{aligned} E(X) &= \frac{r}{p} \\ \text{Var}(X) &= \frac{r(1-p)}{p^2} \end{aligned}$$

Example 18.2. What is another name for a NegativeBinomial($1, p$) distribution?

Example 18.3. Xavier bets on red on roulette until he wins 3 bets and then he leaves. After Xavier leaves, Zander arrives and bets on black until he wins 2 bets and then he leaves. Let X be the number of bets that Xavier makes and let Z be the number of bets that Zander makes.

1. Identify the distribution of X and the distribution of Z .

2. Are X and Z independent? Explain.
 3. What does $X + Z$ represent in this context? Identify the distribution of $X + Z$ without doing any calculations.
 4. Suppose that Yolanda starts making bets on the same game as Xavier, and she bets on black until she wins 5 bets. Does $X + Y$ have a Negative Binomial distribution? Explain.
 5. Suppose instead that Xavier bets on the number 7 until he wins 5 bets. Does $X + Z$ have a Negative Binomial distribution in this case? Explain.
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- If X and Y are independent, X has a NegativeBinomial(r_X, p) distribution, and Y has a NegativeBinomial(r_Y, p) distribution then $X + Y$ has a NegativeBinomial($r_X + r_Y, p$) distribution.
 - A NegativeBinomial(1, p) distribution is a Geometric(p) distribution.
 - If $X_1, X_2, \dots, X_r$ are independent each with a Geometric(p) distribution, then $X_1 + \dots + X_r$ has a NegativeBinomial(r, p) distribution.

In a Negative Binomial situation, the number of successes is fixed so what's random is the number of failures and hence the number of trials. Unfortunately, there is not a standard definition of Negative Binomial in terms of what is counted: failures or trials. It's a good idea to always check the documentation of a software package to see what definition is used.

- Perform Bernoulli(p) trials until a fixed number, r , of successes and then stop. Let X count the total number of *failures* (excluding the r successes). The distribution of X is defined to be the **Pascal(r, p) distribution**.
 - The possible values of a NegativeBinomial(r, p) distribution are $r, r + 1, r + 2, \dots$
 - The possible values of a Pascal(r, p) distribution are $0, 1, 2, \dots$
- If X has a NegativeBinomial(r, p) distribution then $(X - r)$ has a Pascal(r, p) distribution.
- Similarly, if Y has a Pascal(r, p) distribution then $(Y + r)$ has a NegativeBinomial(r, p) distribution.
- Some software packages call “Negative Binomial” what we are calling “Pascal”.